INERTIAL MEASUREMENT, ARTIFACT REJECTION, AND MOTION CONTEXT TELEMETRY

Inertial Context Sensing Overview

The wearable longitudinal penile physiology monitoring device may incorporate one or more inertial measurement units (IMUs) configured to provide motion and orientation context for interpretation of physiologic measurements. Inertial sensing may improve signal reliability by distinguishing physiologic changes from user movement, repositioning events, or environmental disturbances occurring during device operation.

The IMU may include accelerometers, gyroscopes, magnetometers, or combinations thereof, implemented as single-axis, multi-axis, or multi-sensor inertial assemblies. Orientation and motion data may be sampled continuously, periodically, or in event-driven bursts depending on operational mode and power constraints.

Artifact Rejection and Signal Validation

Inertial sensing outputs may be used to identify motion artifacts that could otherwise distort rigidity, pressure, optical, or temperature measurements. Detected motion states may be used to gate sensor sampling, weight computational interpretations, discard unreliable data segments, or trigger adaptive filtering algorithms. Artifact rejection may improve accuracy of longitudinal metrics including erection onset timing, rigidity stability assessment, and nocturnal episode characterization.

Posture and Sleep-State Classification

Orientation and acceleration data may enable classification of user posture and activity state, including standing, seated, supine, lateral, or dynamic movement conditions. During sleep periods, low-motion windows may be identified and correlated with nocturnal erection events to produce enhanced physiologic characterization. Sleep-state segmentation may support analysis of event frequency, duration, and timing distribution across rest cycles.

Optional Movement Telemetry Characterization

In certain embodiments, inertial sensing may additionally enable characterization of rhythmic or repetitive motion patterns associated with user activity. Motion telemetry may be used to derive movement consistency metrics, intensity proxies, or temporal pattern descriptors when permitted by user configuration and privacy settings. Telemetry processing may occur locally or within associated computational systems.

Movement telemetry may be combined with rigidity, perfusion, or temperature sensing outputs to generate contextualized physiologic interpretations reflecting combined mechanical and physiologic states. Such integration may enable improved episode segmentation, behavioral context inference, and longitudinal performance characterization.

Adaptive Sampling Control

Inertial measurements may further control adaptive sampling strategies. Stable conditions may permit reduced sampling rates to conserve energy, while detected motion transitions or physiologic events may trigger temporary increases in sensor acquisition rate. Adaptive sampling may support multiple operating modes balancing fidelity and battery longevity.

The embodiments described herein are illustrative and non-limiting and encompass future inertial sensing technologies and motion interpretation methods capable of improving longitudinal physiologic monitoring accuracy.